

Lucy Robinson

MECHANICAL ENGINEERING (MENG) • THEMED ENTERTAINMENT & RIDE SYSTEMS

📍 Sheffield & London, UK | 📞 +44 7491 232212 | ✉️ lucyalicerob@gmail.com | 🌐 lucyrobinson.engineering | 🔗 [LinkedIn Profile](#)

Royal Academy of Engineering Leaders Scholar [RAEng ELS] and **1st Class MEng Mechanical Engineering** student at the University of Sheffield with hands-on transit rolling stock placement experience at Transport for London (TfL DLR). Combines rigorous structural calculations, SolidWorks mechanism design, and life-safety standards (ASTM F24 / EN 13814) with creative showmanship to engineer high-capacity dark rides, precision turntables, and roller coaster bogies.

🎓 EDUCATION & ACADEMIC HONORS

University of Sheffield — MEng Mechanical Engineering with a Year in Industry

2024 – 2029

1st Class Honours (Years 1 & 2)

- **Honors & Awards:** Royal Academy of Engineering Leaders Scholar (RAEng ELS), Sheffield Engineering Leadership & Service Award (SELSA).
- **Relevant Modules:** Mechanics of Solids, Kinematics & Dynamics, Machine Elements, Fluid Dynamics, Thermodynamics, Mechatronics & Arduino, Materials Science & Manufacturing, Computational Engineering (MATLAB & Python).

Birchwood High School — Bishop's Stortford

2017 – 2024

- **A-Levels:** Mathematics (A*), Further Mathematics (A), Physics (A), German (B) — *Intermediate German Proficiency*.
- **GCSEs (10 Grade 9–8):** Including Maths (9), Physics (9), Chemistry (9), Biology (9), German (9), Geography (9), Design & Technology (8), FSMQ Additional Maths (A).

🏗️ THEMED ENTERTAINMENT & CAD ENGINEERING PROJECTS

Smoke & Starlight AGV — Omnidirectional Trackless Dark Ride

SolidWorks 3D • Python • FEA

2025 – 2026

- Designed a compact omnidirectional ride chassis in SolidWorks featuring independent dual swerve-drive wheel pods for 360° yaw rotation and simulated celestial drift.
- Programmed trajectory kinematic solvers in Python (NumPy/Matplotlib) to compute angular acceleration and G-force envelopes, ensuring passenger compliance with **ASTM F24 standards**.
- Validated structural chassis safety under emergency deceleration and collision load cases using **SolidWorks Simulation (Linear Static FEA)** (Safety Factor > 3.5).

Time Machine Turntable — Subterranean Spatial Illusion Switch Track

SolidWorks • MATLAB • FMEA

2025 – 2026

- Engineered a motorized rotary switch track mechanism utilizing a slewing ring bearing and positive shot-pin locking system to re-align vehicles during vertical drop transitions.
- Conducted first-principles torque, gear reduction, and rotational inertia sizing in MATLAB to achieve smooth 180° rotation in under 4.0s with sub-millimeter indexing repeatability.
- Developed Failure Modes & Effects Analysis (**FMEA**) to ensure fail-safe mechanical interlocks upon loss of electrical power or pneumatic pressure.

The Smugglers Mine Train — Articulated 3-Wheel Coaster Bogie

SolidWorks • Hand Calcs • EN 13814

2025 – 2026

- Designed an articulated 3-wheel bogie cluster (road, side-friction, and upstop wheels) with dual-axis spherical knuckles to eliminate rail-binding on tight mountain switchbacks.
- Calculated beam bending, pin shear, and dynamic wheel load distributions under maximum lateral/vertical G-load cases compliant with **EN 13814 / ASTM F24 standards**.

Assistive Reaching Aid & Para Ice Hockey Equipment — Hackcessible

2024 – 2026

SolidWorks • BS 8888 GD&T • Prototyping

- Led a multidisciplinary engineering team to design, prototype, and present custom assistive mechanical devices for individuals with limited mobility.
- Produced complete manufacturing drawing packs to **BS 8888 standards** with Geometric Dimensioning & Tolerancing (GD&T) and tolerance stack-up analysis.

ENGINEERING & PROFESSIONAL EXPERIENCE

Transport for London (TfL) — Docklands Light Railway (DLR) — London, UK

Summer 2026 – Present

Mechanical Engineering Summer Intern — Electrical & Mechanical Engineering

- Engineered thermal and HVAC load calculation models for depot facilities and station building renovations to support fleet operations.
- Conducted lighting and safety compliance audits across passenger station networks to ensure strict life-safety and emergency egress standards.
- Collaborated with senior rolling stock and infrastructure engineers, gaining deep perspective on preventive maintenance and high-reliability systems.

SELSA Leadership Programme — Faculty of Engineering — Sheffield, UK

Sep 2025 – Present

- Selected for competitive leadership programme; leading the **Vulcan to the Sky Trust** STEM engagement project for the 'Get Up To Speed' fair (March 2026).
- Engaged in industry networking, ethical engineering leadership workshops, and multidisciplinary problem-solving challenges.

36th Sheffield Scout Group — Sheffield, UK

Nov 2025 – Present

Beaver Scout Leader

- Plan and deliver safe, engaging weekly STEM, teamwork, and outdoor activities for 20+ children aged 6–8, fostering curiosity and resilience.

England Golf & Golf Foundation — United Kingdom

2022 – 2024

National Young Ambassador • Recipient of the National Hero's Handshake Award

- Represented junior golfers nationally at PGA and outreach events; promoted Equality, Diversity & Inclusion (ED&I) through the 'Golf Girls Rocks' initiative.
- Authored articles for national publications and attended strategic advisory workshops on youth engagement.

Cantate Youth Choir — Bishop's Stortford & London

2014 – 2024

- 10-year committed ensemble member, performing in high-profile concerts and competitive choral festivals across the UK.

TECHNICAL TOOLKIT & COMPETENCIES

SOFTWARE & COMPUTATIONAL TOOLS

SolidWorks 3D (Assemblies, Mates, FEA Simulation), Autodesk Fusion 360, Autodesk Inventor, AutoCAD, DraftSight, Python (NumPy, Matplotlib), MATLAB & Simulink, Arduino IDE.

MECHANICAL & RIDE ENGINEERING

Dynamic Bogie Kinematics, Mechanism Articulation, First-Principles Stress Sizing, GD&T (BS 8888 / ASME Y14.5), FMEA / Safety Margins, ASTM F24 & EN 13814 Awareness, COTS Sizing (SKF Bearings, Festo Pneumatics).

WORKSHOP & FABRICATION

Rapid Prototyping (FDM / SLA 3D Printing), Manual Lathe & Milling, Precision Drilling, Sheet Metal Bending & Laser Cutting, Fastener Torquing & Metrology (Verniers/Micrometers).

LANGUAGES & LEADERSHIP

German: Intermediate Proficiency (A-Level Qualification) • **Leadership:** RAEng Scholar, SELSA, Project Management, Technical Documentation, STEM Mentoring.